Ideation Phase — Brainstorm & Idea Prioritization

Date	_____
Team ID	_____
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Step 1: Team Gathering & Problem Selection

The team gathered to address the core challenge: EV drivers and fleet managers lack an intuitive visualization tool for charge and range data. The selected problem statement is PS-1: making real-world EV performance data accessible and visual.

Step 2: Brainstorming & Idea Listing

Team brainstormed 8 potential visualization and analysis ideas:

- Interactive geographic range heatmap
- Charge curve: battery % vs. time
- Model comparison for real-world range
- Weather impact (temperature vs. range)
- Terrain effect on energy consumption
- Battery degradation over charge cycles
- Cost analysis per mile by charger type
- Trip planner with smart charge stops

Step 3: Idea Prioritization

#	Idea	Category	Feasibility (H/M/L)	Priority
1	Interactive range heatmap on a geographic map	Visualization	High	High
2	Charge curve chart (battery % vs. time per session)	Visualization	High	High
3	Range comparison by vehicle model	Analysis	High	High
4	Temperature impact on range (cold/hot weather)	Analysis	Medium	High
5	Terrain effect on energy consumption	Analysis	Medium	Medium

6	Battery degradation trend over time	Trend Analysis	Medium	Medium
7	Cost-per-mile by charge source (home/public/fast)	Cost Analysis	Low	Medium
8	Trip planner with optimal charge stop suggestions	Feature	Low	Low

Selected Ideas for Implementation

Based on feasibility and priority, the following ideas are selected for Sprint-1 and Sprint-2:

- Range Heatmap (geographic view)
- Charge Curve Chart
- Model Comparison View
- Temperature vs. Range Analysis